

Employee Attrition Prediction using Machine Learning

Executive Summary

This project focuses on predicting employee attrition using Machine Learning techniques. The objective is to identify employees who are more likely to leave the organization so that the Human Resources (HR) team can take proactive steps to improve employee retention.

The IBM HR Analytics dataset containing 1,470 employee records was used for this analysis. The data was cleaned by removing unnecessary columns, converting categorical values into numerical format, and scaling numerical features before building the machine learning models.

Three classification models were developed and compared:

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

The analysis showed that employee attrition is influenced by multiple factors rather than salary alone. Important factors include Monthly Income, Overtime, Age, Total Working Years, Work-Life Balance, and Years at the Company.

The exploratory data analysis revealed that the Research & Development and Sales departments experience higher employee attrition compared to other departments. Certain job roles such as Sales Executive, Laboratory Technician, and Research Scientist also show relatively higher employee turnover.

Based on these findings, HR should focus on improving employee work-life balance, reviewing compensation for high-risk employees, providing career growth opportunities, and conducting regular employee engagement programs.

Although the machine learning model provides valuable predictions, it should be used as a decision-support tool. Personal circumstances, future organizational changes, and external economic factors are not included in the dataset and may also influence employee attrition.

Overall, this project demonstrates how Machine Learning can support HR teams in making data-driven decisions that improve employee retention and reduce organizational costs.

